

9.5/15



ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY

College of -----CNSS-----

Department of FRESHMAN ENGINEERING

Mathematics for Natural Sciences (Math 1007)

Test-I

Date: August, 2021

Time allowed: 45 min

Maximum Score 15 %

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Part I: Write "True" if the statement is correct and "False" if the statement is Wrong on the space provide (5 points)

True 1. If $p \Leftrightarrow q$ and $\neg p$ are true, then q is false

False 2. If $A \subseteq B^c$, then $A \cap B = \phi$.

False 3. If $A \subseteq B$, then $A^c \subseteq B^c$

False 4. $(\exists x \in \mathbb{R})(\forall y \in \mathbb{R})(x^2 + yx = y + 1)$

True 5. Every composite number can be expressed as a product of its prime factors. and this factorization is unique except the order of the factors .

Part II: Short Answer Questions (5 points)

Write the Most simplified answer in the space provided

1. The GCF of two numbers is 3 and their LCM is 180. If one of the numbers is 45, then the second number is 12

2. Let $A = \{ \phi, a, c \}$ and $B = \{ \{ \phi \}, c \}$, then $A \cap B =$ C

3. Let $A \cup B = \{0,1,2,3,4,5\}$, $A \cap B' = \{0,1\}$ and $A \cap B = \{2,4\}$. Then

$B \setminus A =$ ϕ

4. The fraction form of $3.\overline{25}$ is $\frac{293}{990}$

5. The argument of the complex numbers $z = \frac{3i}{-1-i}$ is $\tan^{-1}(1) = \pi/4$

Part III: Workout Questions [5 pts]

1. Use truth table method to determine the validity of the argument:

$$p \Rightarrow q, \neg r \Rightarrow \neg q \vdash \neg r \Rightarrow \neg p$$

P	q	r	$\neg r$	$\neg q$	$\neg p$	$p \Rightarrow q$	$\neg r \Rightarrow \neg q$	$\neg r \Rightarrow \neg p$	C
T	T	T	F	F	F	T	T	T	T
T	T	F	T	F	F	T	F	F	F
T	F	T	F	T	F	F	T	T	T
T	F	F	T	T	F	F	T	F	F
F	T	T	F	F	T	T	T	T	T
F	T	F	T	F	T	T	F	F	F
F	F	T	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T	T	T

So it is **VALID** b/c when the premises are True the conclusion also true

2. Find the square roots of the complex number $z = 9i$

[3 pts]

$$r = 9$$

$$\theta = \tan^{-1}\left(\frac{9}{0}\right) = \text{Undefined}$$

$$\theta = \pi/2$$

$$z = re^{i\theta}$$

$$z_0 = r^{1/n} e^{i\left(\frac{\theta}{n} + \frac{2\pi k}{n}\right)}$$

$$z_0 = 9^{1/2} e^{i\left(\frac{\pi/2}{2} + \frac{2\pi k}{2}\right)} \quad k = -1, 0, 1$$

when $k=0$

$$z_0 = 9^{1/2} e^{i(\pi/4)}$$

$$= 3e^{i(\pi/4)}$$

$$z_0 = r(\cos\theta + i\sin\theta)$$

$$= 3\left(\cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right)\right)$$

$$z_0 = \frac{3\sqrt{2}}{2} + \frac{3\sqrt{2}i}{2}$$

when $k=1$

$$z_0 = 9^{1/2} e^{i\left(\frac{\pi}{4} + \pi\right)}$$

$$= 3e^{i(5\pi/4)}$$

$$z_0 = r(\cos\theta + i\sin\theta)$$

$$= 3\left(\cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4}\right)$$

$$z_0 = -\frac{3\sqrt{2}}{2} - \frac{3\sqrt{2}i}{2}$$